Scale-Aware Adaptive Refinement and Cross-Interaction for Remote Sensing Audio-Visual Cross-Modal Retrieval

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***Abstract*— Remote sensing (RS) audio-visual cross-modal retrieval is a challenging task in the search of meaningful RS information. Nevertheless, the impact of multiscale features and associated redundant information in the RS images cannot be overlooked in the retrieval task. In addition, how to deal with the completely different physical expressions of different modal information is crucial for cross-modal retrieval tasks. To tackle these issues, we propose a Scale-aware Adaptive Refinement and Cross-Interaction (SARCI) network. The Quaternion-attention Dominated Multiscale Visual Refinement (QDMVR) module in SARCI is suggested to learn multiscale visual features and further optimize features containing redundant information for different scale features. To better integrate channel attention and spatial attention for adaptively learning of meaningful visual semantics, we propose the symmetric quaternion attention (SQA) within the QDMVR module to enhance RS visual features. The SQA mechanism acts on both high-level and low-level features to explore salient RS vision information across different scales. In order to allow information from different modalities to interact more valuably, we propose the Instruction-based Cross-Learning Module (ICLM) to perform cross-modal feature interaction based on the characteristic of the two modalities. SARCI network demonstrates state-of-the-art performance on three public RS cross-modal datasets: Sydney, UCM, and RSICD audio-visual datasets. The code is available at: https://github.com/WUTCM-Lab/SARCI.**

***Index Terms*— Cross-modal feature interaction, remote sensing (RS) audio-visual cross-modal retrieval, scale-aware adaptive refinement and cross-interaction (SARCI).**

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I. Introduction

OVER the past years, many researchers have been drawn to the wealth of information offered by remote sensing (RS) technology. RS technology holds a crucial role in earth observation, environmental monitoring, and natural resource management [1], [2], [3], [4], [5]. As the amount of RS data continues to increase, we are faced with a greater challenge: how to manage, analyze, and retrieve these massive data more effectively to obtain meaningful information from them. As an important cross-technical field, research on automatic RS information retrieval methods has been well developed.

The development of RS information retrieval has gone through the stages from single-modal RS image retrieval to multimodal RS information retrieval. The initial rise of RS image retrieval refers to the use of RS images to retrieve related RS images in the RS image database [6], [7], [8], [9]. These traditional image retrieval methods definitely rely on visual information. However, these methods cannot meet the increasingly diverse RS data needs in the real world. Recently, there has been a growing focus on researching the utilization of multiple modalities of RS information for retrieval tasks. Some researchers are committed to building cross-modal RS information datasets, including annotating other modal information on existing RS image datasets [10], [11] or building more massive datasets [12], [13], [14] to promote advancements of multimodal RS information retrieval. Researchers use annotated RS text description information or audio information and RS images to implement cross-modal RS information retrieval. Image-text cross-modal RS information retrieval [12], [15], [16], [17], [18], [19] and audio-visual cross-modal RS information retrieval [10], [11], [20], [21], [22], [23], [24]

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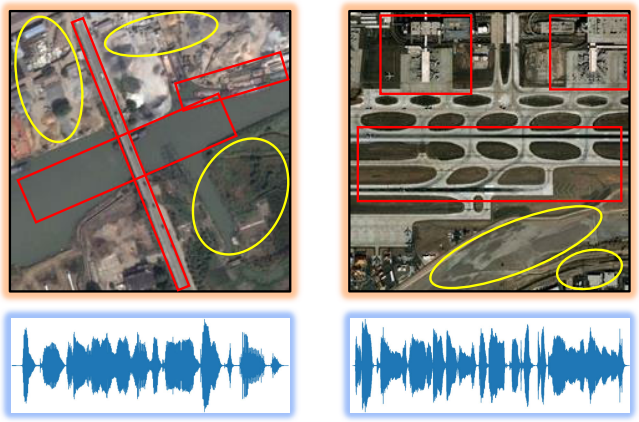


Fig. 1. Two data examples of RS image-audio pairs. We utilize red rectangles to outline the objects that need to be identified, while circling areas containing redundant information. (a) Information of audio includes three scales of salient objects corresponds to the RS image. While some redundant information (e.g., green parts along the river, buildings) in the image may misguide the retrieval model. (b) Small-scale planes, buildings, and two parallel runways are RS objects to be identified, while redundant things in the yellow circle need to be filtered.

have become important trends in the analysis, utilization, and exploration of RS data. These studies not only utilize visual data but also combine text description information or audio information to provide a more comprehensive and diverse perspective on data application. In this article, our research delves into the exploration of audio-visual cross-modal RS information retrieval. In practical applications, humans can directly speak the desired RS information to perform retrieval tasks.

Lately, there have been introductions of end-to-end methods for audio-visual cross-modal RS retrieval [10], [11], [20], [21], [22], [23], [24]. However, the field still faces some important challenges. First of all, the multiscale characteristics of RS images have not been fully utilized. Accompanying the multiscale characteristics are the prominent information redundancy phenomena (examples as shown in Fig. 1) of RS images in retrieval tasks. On the one hand, utilizing visual information across multiple scales is necessary to achieve better retrieval results. On the other hand, the model needs to adaptively learn to capture the salient information from the complex RS scenes. We observe that in [11] and [12], the handling of multiscale visual features and redundant visual information is split into two stages. This approach is somewhat simple and lacks careful consideration of the visual scale characteristics where redundant information may be present. To address this problem, we propose the Quaternion-attention Dominated Multiscale Visual Refinement (QDMVR) module to adaptively explore the salient RS information within different scale visual features.

Second, a significant challenge in cross-modal retrieval is the huge difference in expression of various modalities. Features of different modalities have completely different physical expressions. How to better embed different modal features in public space and accurately learn the correlation between different modalities is the key. Fig. 1 presents two samples of RS image–audio pair that requires a model to learn informative correspondences across the modalities. In related research on cross-modal features, performing meaningful cross-modal interactions or cross-modal alignment is the mainstream method for learning correlations between different modalities [25], [26]. Additionally, the interaction between modalities in previous study (e.g., [11]) is easily implemented to a certain degree and does not take into account the characteristics of audio and visual information. In the article, we build an Instruction-based Cross-Learning Module (ICLM) to explore the complex relationships between modalities and enhance the understanding of the accurate RS semantic correspondences.

Additionally, most of the previous audio-visual cross-modal RS retrieval research used simple audio encoders [11], [22], [24] such as ResNet18. We found that they did not fully utilize the information of audio feature. Based on the mainstream voice network Emphasized Channel Attention, Propagation, and Aggregation in Time Delay Neural Network (ECAPA-TDNN) [27] in the speaker recognition field, we utilize an improved audio encoder Enhanced-ECAPA-TDNN (E-ECAPA-TDNN) for the first time in response to the problem of insufficient audio feature encoding capabilities.

Overall, our contributions are as follows.

1) Taking into account the complex features of RS images with multiscale and redundant information, we propose a novel QDMVR module. We aim to fully extract multiscale information while adaptively learning salient information across scales to enhance the representational capacity of visual features.
2) In order to overcome the expression differences between different modalities, we introduce the ICLM to perform cross-modal feature interactions. The ICLM can effectively promote information in different modalities to instruct each other, allowing the model to learn the correlation between information in different modalities more effectively.
3) Extensive experiments on three common public audio-visual cross-modal RS datasets: Sydney, UCM, and RSICD, demonstrating state-of-the-art performance of the proposed Scale-aware Adaptive Refinement and Cross-Interaction (SARCI) model.

The subsequent article is structured as follows. We provide a brief summary of some prior related research on RS information retrieval in Section II. In Section III, all proposed modules and strategies in our SARCI model are introduced. In Section IV, we describe the performed experiments, presenting the detailed design of the comparison and ablation experiments as well as the experimental results. We perform a comprehensive set of quantitative and qualitative experiments to evaluate the effectiveness of the entire SARCI model and its constituent modules. In Section V, we conclude the article in general.

II. Related Work

In this segment, we review the evolution of RS information retrieval, spanning from the early stages of RS image

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retrieval to the more advanced realms of cross-modal image-text retrieval and audio-visual retrieval.

A. RS Image Retrieval

Due to the rapid advancements in RS sensor technology, there has been a surge in the volume of RS images. Effectively retrieving and managing this extensive data has become an imminent issue. In the domain of RS image retrieval, the content-based image retrieval (CBIR) method has gained prominence due to its outstanding performance. To precisely characterize image similarity in RS Image Retrieval, Liu et al. [6] introduce an approach that concurrently learns deep image features and precise image similarity, thereby improving image retrieval performance by incorporating latent relationships between images.

Many current retrieval methods focus on global image features and struggle to differentiate between semantically similar images with distinct visual representations. In response, Huang et al. [7] introduce a supervised contrastive learning approach based on the fusion of global and local features, enhancing image expression capabilities. The incorporation of supervised contrastive learning improves feature distribution in retrieval tasks. To extract representative features from RS images and enhance large-scale CBIR efficiency, Liu et al. [8] capture intricate RS image content and transform features into hash codes.

Training a CNN typically demands abundant labeled samples, yet RS images with labels are often limited. Therefore, Liu et al. [9] transform similarity learning into deep ordinal classification. The model can be trained from scratch using entirely unlabeled RS data, leveraging knowledge transferred from everyday image sets. Neglecting multiscale and saliency information in RS images may result in feature loss and hash code balance challenges. In response, Zhao et al. [28] propose to obtain finer multiscale features and larger receptive fields through merged multiscale residual blocks.

With the massive abundance of RS information, cross-modal RS retrieval methods have gradually emerged because their input forms are relatively flexible and can leverage RS data to its fullest extent.

B. Image-Text RS Retrieval

Research focused on image-text cross-modal retrieval methods has garnered significant attention. Integrating a multiscale visual self-attention (MVSA) module, AMFMN [12] extracts salient features from RS images. AMFMN also designs guidance modes between modalities. Moreover, Yuan et al. [12] curate RSITMD, a fine-grained and more challenging dataset tailored for the task. Introducing a lightweight multiscale cross-modal retrieval framework, Yuan et al. [15] incorporate knowledge distillation and semi-supervised optimization.

Presenting a novel framework, Yuan et al. [16] optimize both the representation matrix and adjacency matrix of local features. This approach integrates multilevel features, culminating in superior retrieval results. Treating RS images and texts collectively, Tang et al. [17] concurrently process them, facilitating the effective learning of multimodal features and the direct alignment of their cross-modal semantic relationships. Zhang et al. [18] design a curriculum learning-based adaptive alignment strategy that aligns image-text pairs from easy to difficult. Zhang et al. [19] introduce a mask-guided attention strategy for learning salient regions and words to enhance the model's unimodal learning ability and eliminate unnecessary and redundant information from each modality.

C. Audio-Visual RS Retrieval

The research on audio-visual RS retrieval is categorized into two main approaches: retrieving images or audios of corresponding categories and retrieving paired images or audios. Most research like [10], [20], [21], [22], [23], [24] focuses on investigating the former. In order to achieve more precise retrieval and advance the development of this field, Huang et al. [11] initiate the study of models capable of retrieving paired images or audios. So, we also conduct research on retrieving paired images or audios in this article.

Originally, Guo et al. [10] introduce a novel deep network designed to directly capture the correspondence across the images and audios. In the study by Guo et al. [20], the semantic relation between image and audio serves to supervise the training process. Additionally, the corresponding RS images are employed as supervision information to capably explore key features of spoken audio.

An innovative method presented by Chen et al. [21] leverages the congruity of deep features and multiscale information to streamline the acquisition of low-dimensional binary representations. Presented by Ning et al. [22], the method for audio-visual RS retrieval stands out for its unique approach. The method effectively takes into account pairwise, intra-modal, and non-pairwise inter-modal relationships simultaneously, thereby improving the semantic consistency of the learned representations.

The approach proposed by Chen et al. [23] introduces a novel technique for generating hash codes by capturing the relative semantic similarity relationships and category-level semantic information within cross-modal data. The model involves the creation of a robust module that utilizes a unique quadruple formulation to identify and incorporate relevant samples, facilitating the learning process for relative semantic similarity relationships. Chen et al. [24] propose a method to capture distinctive information and utilize precise alignment to learn hash codes for RS images and audio.

As a groundbreaking research for retrieving paired images or audios, Huang et al. [11] introduce to combine multiscale feature extraction with audio-driven reduction of redundant information in images. This method focuses on seizing distinctive discriminative regions through the extraction of localized details to enhance the overall global modeling process.

We find that the method in [11] divides the processing of multiscale visual feature and visual redundant information into two stages. This processing method is relatively rough and does not carefully consider the visual scale characteristics, where redundant information may appear. Therefore, we propose the QDMVR module to adaptively process visual redundant information at different scales. In this way, possible redundant information can be adaptively removed in detail

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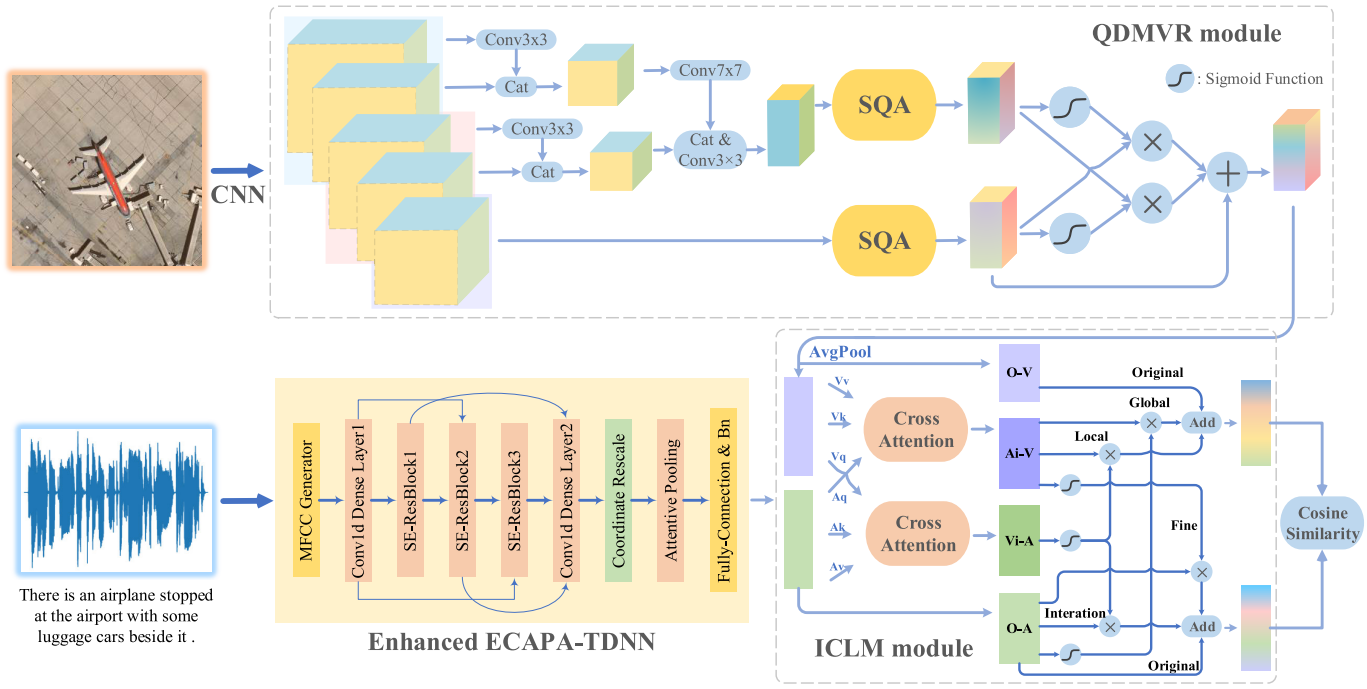


Fig. 2. Whole architecture of the SARCI network. The overall structure consists of four main parts. The CNN backbone ResNet18 is applied to encode RS visual information. The E-ECAPA-TDNN processes the RS audio input as audio feature for further cross-modal interaction. The QDMVR module tackle the RS visual characteristics of multiscale and redundant information by detailed refinement with the SQA mechanism. Finally, the ICLM is responsible for meaningful interactions between cross-modal features. Within the illustration of the ICLM, the symbols "O-V," "Ai-V," "Vi-A," "O-A" denote the features V_o, V_{Ai}, A_{Vi}, A_o respectively.

and more salient visual retrieval features can be obtained. Moreover, the interaction between modalities in [11] is conducted simply to some extent and is not designed based on the characteristics of audio and visual information. To this end, we propose the ICLM to perform more efficient cross-modal interactions. Furthermore, many previous audio-visual RS retrieval methods [11], [22], [24] utilize relatively simple audio encoder, which greatly limits the performance of the retrieval system. For this issue, we design an improved audio encoder E-ECAPA-TDNN for the audio-visual RS retrieval task.

III. Proposed Methodology

The general view of SARCI model is illustrated in Fig. 2 with four parts. The RS visual encoder pretrained ResNet18, audio encoder E-ECAPA-TDNN, QDMVR module, and ICLM form the whole SARCI network, which trains with the audio-visual hinge-based triplet ranking loss [29], [30], [31]. The RS images are encoded using the CNN backbone ResNet18. The RS audio input is processed by the E-ECAPA-TDNN to extract audio features. The QDMVR module addresses the multiscale and redundant visual characteristics of RS visual input through detailed refinement, generating comprehensive visual feature representation. The cross-modal interaction module, ICLM, generates corresponding cross-learned features of a couple of audio-visual input features fed by the E-ECAPA-TDNN and the QDMVR module. The cosine similarity of one couple of cross-learned features leads the inference process, calculating the metrics for the whole model.

A. Problem Formulation

For RS audio-visual retrieval, the query could be an RS image, and the objective is to retrieve the most pertinent audio(s) from a database. Conversely, the query might be a piece of audio, with the aim to retrieve relevant RS image(s). This involves two subtasks, designated as audio retrieval and image retrieval, respectively. The primary objective is to improve recall at K ($R@K$), where K represents the number of top items returned. $R@K$ represents the fraction of queries, where the most pertinent item is ranked among the top K items returned.

Let $T = \{(\mathcal{I}_n, \mathcal{S}_n)\}_{n=1}^{N}$ represent a training set comprising N RS image–audio pairs. Herein, $(\mathcal{I}_n, \mathcal{S}_n)$ denotes positive pairs, whereas $(\mathcal{I}_n, \mathcal{S}_{m\neq n})$ signifies negative pairs. Accordingly, $\mathcal{S}_n$ is construed as the most pertinent audio counterpart to RS image $\mathcal{I}_n$, and conversely. A similarity function $s(\mathcal{I}, \mathcal{S})$ is defined to discern positive pairs with greater affinity compared to negative ones. In the realm of audio retrieval, the query is an image, and the ranking of an audio database is contingent upon this similarity function. The metric $R@K$ quantifies the proportion of queries wherein the positive audio is ranked within the top K audio candidates based on $s(\mathcal{I}, \mathcal{S})$. Analogously, this metric applies to image retrieval scenarios. In this article, we apply cosine similarity between the cross-modal features explored by the whole network

$$(V, A) = \phi\{(\mathcal{I}, \mathcal{S}); \theta_\phi\} \tag{1}$$

where ϕ represents the whole SARCI network to map the image–audio input $(\mathcal{I}, \mathcal{S})$ into cross-learned features (V, A), and θ_ϕ denotes the parameters within the network. In specific,

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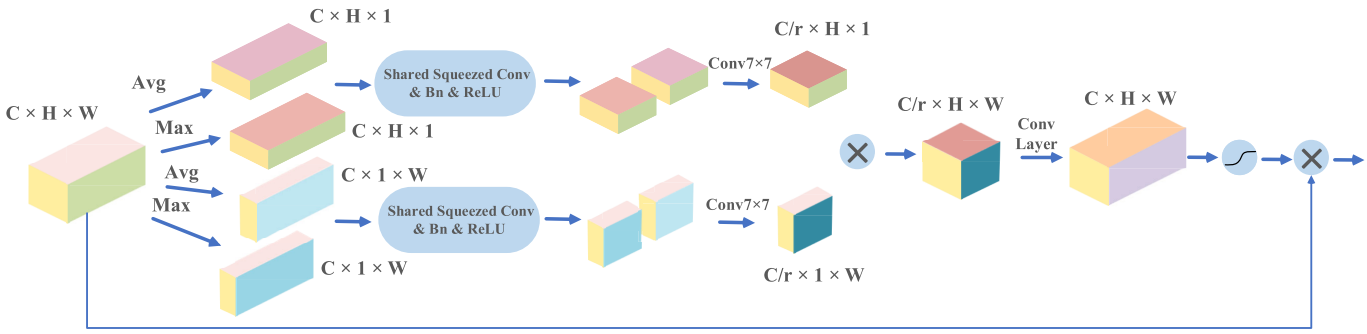


Fig. 3. Procedure of the SQA mechanism. Quaternary branches symmetrically shape the main architecture. The symbols "Avg" and "Max" denote the operations of average pooling and maximum pooling respectively.

the final visual interaction feature V and the final audio interaction feature A are constructed through (25) and (28), respectively.

Then, the cosine similarity between the modalities is calculated as

$$s(\mathcal{I}, \mathcal{S}) = cs(V, A) = \frac{V \cdot A}{\|V\|\|A\|} \tag{2}$$

where cs denotes the cosine similarity function for (V, A). $\cdot$ represents the dot product. $\|V\|$ and $\|A\|$ represent the norms of V and A, respectively.

B. Visual and Audio Encoder

1) Visual Encoder: For the purpose of restricting parameter count and preventing possible overfitting issues, we utilize the ResNet18 [32], which was pre-trained on ImageNet to generate RS image representations. The outputs of five posterior layers in this CNN network are transferred into the latter module for further exploring the meaningful visual semantics

$$F^i = \text{ResNet}(\mathcal{I}), i = 1, 2, \ldots, 5 \tag{3}$$

where i represents the layer index of the ResNet network.

2) Audio Encoder: ECAPA-TDNN [27] was first proposed to serve as the speaker embedding extractor of the speaker verification task within the field of audio processing. In ECAPA-TDNN, the incorporation of 1-D squeeze-excitation blocks, multiscale Res2Net features, extra skip connections, and channel-dependent attentive statistics pooling made an extraordinary feature extractor for speaker verification. In this article, we exploit it for encoding audio representation in our audio-visual RS retrieval task. As illustrated in Fig. 2, our E-ECAPA-TDNN network rescales the audio feature by coordinate rescale layer after the dense layer, which processes the concatenated information of prior there SE-Res2Blocks. The coordinate rescale layer is consisted of coordinate attention [33], which was proposed for reducing spatial information loss when training the network. Similarly, the audio feature could be considered to reduce temporal context information loss when concentrating them by attention module. We remain the 1-D SE block in prior three SE-ResBlocks out of the model overfitting risk. The coordinate rescale layer generally optimizes the concatenated features by wiping out the unimportant information and concentrating on the salient retrieval clues when interacting with other modal feature.

C. Quaternion-Attention-Dominated Multiscale Visual Refinement

In RS cross-modal retrieval field, object detection techniques can help to obtain the salient object context. However, the multiscale character in RS images is worth exploring for the cross-modal RS retrieval task. In particular, in the face of the latter interaction with another modality, adequate and appropriate visual information extracted from RS images matters for the task. As shown in Fig. 2, the QDMVR module is designed to learn comprehensive visual features for the RS images.

1) SQA Mechanism: In recent years, plenty of attention mechanisms have arisen on account of salient information extracting or feature strengthening for almost all directions utilizing deep learning algorithms. Distinguished attention research, such as squeeze-and-excitation (SE) block [34] and convolutional block attention module (CBAM) [35], are suggested to apply on various mainstream tasks for better performance. SE block is early suggested to use channel attention for feature enhancement. Later, the CBAM mechanism is recognized for its sequential combination of channel attention and spatial attention to enhance the processed information. Recently, coordinate Attention [33] mechanism has been proposed for the purpose of protecting spatial information when focusing on channel information. We find that CBAM uses both average pooling and maximum pooling to better represent features. Nevertheless, coordinate attention only performs average pooling along the height and width directions when processing the spatial context.

Inspired by the aforementioned considerations, we propose a new symmetric quaternion attention (SQA) mechanism in our QDMVR module to better integrate channel attention and spatial attention for refining visual features. To adaptively focus on key visual retrieval information, four balanced branches in the SQA mechanism utilize average pooling along with maximum pooling in two directions. Additionally, unlike sequenced steps designed for channel attention and spatial attention in CBAM, the SQA is organized in a fusion mode. As illustrated in Fig. 3, quaternary branches symmetrically shape the main architecture of our proposed SQA mechanism.

The SQA can directly operate on the input features, enhancing their expressiveness for the specified task. The overall operation of the SQA mechanism is as follows. The module

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is plug-and-play and does not change the size of the input feature F

$$F' = \mathrm{SQA}(F) \tag{4}$$

where SQA denotes the SQA mechanism.

Specifically, as shown in (5) and (6), the input feature F is first subjected to average-pooling and maximum-pooling operations along both dimensions of width and height. These four branches obtain relevant spatial variation features in the height and width dimensions, respectively. We record the four features as Fh_α, Fh_μ, Fw_α, and Fw_μ, respectively

$$Fh_\alpha = \alpha_w(F),\ F\ h_\mu = \mu_w(F) \tag{5}$$

$$Fw_\alpha = \alpha_h(F),\ F\ w_\mu = \mu_h(F) \tag{6}$$

where α_w and α_h denote the average pooling along the width and height dimensions, respectively, and μ_w and μ_h denote the maximum pooling along the width and height dimensions, respectively.

The corresponding features of the four branches are squeezed separately, among which Fh_α and Fh_μ share the weight of the squeeze layer. The same operation is done for Fw_α and Fw_μ. The specific steps are as follows:

$$Fh'_\alpha = \mathcal{RL}(\mathcal{BN}(\varphi_1(Fh_\alpha))),\ F\ h'_\mu = \mathcal{RL}\big(\mathcal{BN}\big(\varphi_1\big(Fh_\mu\big)\big)\big) \tag{7}$$

$$Fw'_\alpha = \mathcal{RL}(\mathcal{BN}(\varphi_1(Fw_\alpha))),\ F\ w'_\mu = \mathcal{RL}\big(\mathcal{BN}\big(\varphi_1\big(Fw_\mu\big)\big)\big) \tag{8}$$

where $\mathcal{BN}$ represents the batch normalization [36]. $\mathcal{RL}$ represents the ReLU activation function. φ_1 denotes a convolution layer that squeezes the feature to 1/8 channel size with a kernel size of (1, 1), a stride of 1, and a padding value of 0.

Then, we concatenate Fh'_α, Fh'_μ and Fw'_α, Fw'_μ along channel dimension. The two concatenated features are subsequently fed through a convolution layer with a convolution kernel size of 7×7 in parallel to further fuse features in the width and height dimensions. The convolution layer also squeezes the doubled channel size to half. The process is formalized as

$$Fh = \varphi_7\big(\big[Fh'_\alpha, F\ h'_\mu\big]\big) \tag{9}$$

$$Fw = \varphi_7\big(\big[Fw'_\alpha, F\ w'_\mu\big]\big) \tag{10}$$

where φ_7 denotes the convolution layer mentioned above, with a kernel size of (7, 7), a stride of 1 stride, and a padding value of 3. [,] denotes the channel-wise concatenation function.

Next, we multiply the Fh and Fw above to obtain rich attention information in the spatial dimension as a whole. Then, we further expand the features to the original channel dimensions as F and apply it to the original feature F. In this way, the original input feature F gets optimized attention information

$$F_Q = \tau_1(Fw \otimes Fh) \tag{11}$$

$$F' = \sigma\big(F_Q\big) \otimes F \tag{12}$$

where τ_1 denotes the convolution layer to expand channel dimension with a kernel size of (1, 1), a stride of 1, and a padding value of 0. The symbol σ represents the sigmoid function. And $\otimes$ denotes element-wise product. F_Q denotes the feature of attention information.

2) Details of QDMVR Module: As presented in Fig. 2, we incorporate the SQA mechanism twice into our QDMVR module. The SQA is used to further refine both the low-level visual feature containing smaller scale feature information and the high-level visual feature containing the most significant feature information. Specifically, the salient visual information contained within the visual features will be attended to. The SQA mechanism can explore visual salient information by its attention capability. Actually, the SQA mechanism focuses on useful retrieval information and adaptively processes visual redundant information at different scales.

We utilize the visual features F^i extracted from (3) to generate low-level visual feature F_l and high-level visual feature F_h. We define the process as

$$F^{12} = \big[\tau_3\big(F^1\big), F^2\big] \tag{13}$$

$$F^{34} = \big[\tau_3\big(F^3\big), F^4\big] \tag{14}$$

$$F_l = \tau_3\big(\big[\tau_7\big(F^{12}\big), F^{34}\big]\big) \tag{15}$$

where τ_3 denotes the convolution layer with kernel size (3, 3), 2 strides and padding value 1. τ_7 denotes the convolution layer with a kernel size of (7, 7), a stride of 4, and a padding value of 3. In addition, τ_3 and τ_7 do not change the channel size of features

$$F_h = F^5. \tag{16}$$

Afterward, the low-level visual feature F_l and high-level visual feature F_h are refined by the SQA mechanism

$$F'_l = \mathrm{SQA}(F_l),\ F'_h = \mathrm{SQA}(F_h). \tag{17}$$

In the next stage, we will symmetrically synergize the optimized low-level visual feature and high-level visual feature to learn the final comprehensive visual feature containing multiscale information. This stage is formalized as follows:

$$F_g = \big(\sigma\big(F'_l\big) \otimes F'_h\big) \oplus \big(\sigma\big(F'_h\big) \otimes F'_l\big) \tag{18}$$

where $\oplus$ denotes the vector summation.

Since optimized high-level visual feature F'_h contain more important information for the retrieval task, we apply residual connection then. We add the multiscale fusion information F_g above with the optimized high-level visual feature F'_h to obtain the final multiscale comprehensive visual feature representation V_g.

D. Instruction-Based Cross-Learning Module

Different modality has naturally different physical expressions. Therefore, to achieve cross-modal retrieval, we must strive to learn the corresponding relationship of different modalities to alleviate the existing expression differences. There have been many good attempts at matching information from different modalities. Nevertheless, it still requires detailed design based on specific tasks so that the information between different modalities can be fully utilized. Cross-modal feature interaction plays a crucial role in audio-visual feature relationship learning. With regard to the issue, we propose

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ICLM, which enables cross-modal information to instruct each other and promote the cross-modal interactions. The structure of ICLM is shown in Fig. 2.

First, visual feature V_g obtained by QDMVR is average pooled to have appropriate feature dimensions interacting with audio features. The process is shown in the following formula:

$$V_o = \psi\left(V_g\right) \tag{19}$$

where V_o denotes the pooled visual feature. ψ denotes the average-pooling layer upon the whole spatial area.

In order to enable preliminary interaction between visual and audio information, we use two parallel cross-attention [37] modules to reconstruct the two modal features. Then, audio features instructed by visual information and visual features instructed by audio information are obtained. The audio feature extracted by E-ECAPA-TDNN is defined as A_o. The details are as follows:

$$\begin{aligned} V_q &= \Gamma_{vq}(V_o),\ V_k = \Gamma_{vk}(V_o),\ V_v = \Gamma_{vv}(V_o) \\ A_q &= \Gamma_{aq}(A_o),\ A_k = \Gamma_{ak}(A_o),\ A_v = \Gamma_{av}(A_o) \end{aligned} \tag{20}$$

where Γ denotes the linear layer. Γ_{vq}, Γ_{vk}, and Γ_{vv} denote the linear layer projecting the visual feature V_o into corresponding query, key, and value, respectively. So, the features V_q, V_k, V_v represents query, key, and value for the visual branch. Γ_{aq}, Γ_{ak}, and Γ_{av} denote the linear layer projecting the audio feature A_o into corresponding query, key, and value, respectively. The features A_q, A_k, A_v represents query, key, and value for the audio branch

$$V_{Ai} = \Lambda\left(\text{query} = A_q, \text{key} = V_k, \text{value} = V_v\right) \tag{21}$$

$$A_{Vi} = \Lambda\left(\text{query} = V_q, \text{key} = A_k, \text{value} = A_v\right) \tag{22}$$

where $\Lambda(\text{query}, \text{key}, \text{value})$ denotes the multihead cross-attention function [37]. V_{Ai} and A_{Vi} are visual features instructed by audio information and audio features instructed by visual information, respectively.

As can be seen from the above representation, we perform preliminary feature interaction operations by intersecting the queries of visual information and audio information and then using multihead cross-attention Λ twice in parallel.

Consequently, both modalities after preliminary mutual instruction acquire more clear semantics. For the two modalities, visual information is more complex, while audio information points out relevant semantics straightforwardly. Moreover, we believe that audio features instructed by visual information have a clearer representation of RS information. Therefore, we aim to fully leverage the significance of the feature A_{Vi}. We further conduct detailed instruction to generate the final feature representation of visual and audio branch.

For RS visual information, some redundant information has been eliminated under the instruction of audio salient feature after the preliminary feature cross-learning. Therefore, the visual features after audio instruction serve as the main information to participate in further feature interaction for the visual branch. We believe that the audio feature A_{Vi} after visual instruction can represent salient RS information in a local range, while the original audio feature A_o has more global RS information. Therefore, in the next step of feature interaction, we carried out the following detailed design for the visual branch:

$$V_{\text{local}} = \sigma(A_{Vi}) \otimes V_{Ai} \tag{23}$$

$$V_{\text{global}} = \sigma(A_o) \otimes V_{Ai}. \tag{24}$$

The original audio feature A_o and the audio feature A_{Vi}, which is under the visual instruction, are utilized to further interact with the visual feature V_{Ai}. The local and global interactive features V_{local} and V_{global} of the visual branch are obtained.

We perform the following operations to obtain the final visual interaction feature V:

$$V = V_{\text{local}} \oplus V_{\text{global}} \oplus V_o. \tag{25}$$

The addition of local and global interactive visual features aims to fully characterize visual features in cross-modal tasks. In order to avoid some subtle important visual information being ignored during the interaction process, we deliberately add the residual connection with the visual original feature V_o.

For the audio branch, the encoded audio feature before visual instruction already has significant RS comprehensive information. Therefore, in the next interaction of the audio branch, we mainly use the original audio feature A_o.

The operation of audio instruction for the visual feature V_{Ai} acquaints it with effective cross-modal instruction capability. So, we first utilize V_{Ai} to instruct the original audio feature A_o and learn more fine-grained RS audio feature A_{fine}

$$A_{\text{fine}} = \sigma(V_{Ai}) \otimes A_o. \tag{26}$$

We do not directly deploy the audio feature A_{Vi} after visual instruction. Nevertheless, it is adopted to further instruct the original audio feature A_o. The secondary iteration mode is to explore more suitable retrieval semantics

$$A_{\text{iteration}} = \sigma(A_{Vi}) \otimes A_o. \tag{27}$$

Overall, the fine-grained RS audio feature A_{fine}, the secondary iteration audio feature $A_{\text{iteration}}$, and the original audio feature A_o build up the final audio interaction feature A

$$A = A_{\text{fine}} \oplus A_{\text{iteration}} \oplus A_o. \tag{28}$$

After the above-mentioned sufficient two-stage cross-modal feature interaction, richer and more comprehensive cross-modal interaction information is explicitly obtained for the visual and audio branches in different ways. In the aggregate, the two modalities are interactively supplemented and refined. Such optimization is dedicated to better aligning and matching the information of the two modalities.

E. Objective Function

Multimodal feature alignment has made considerable progress, particularly with the emergence of triplet loss as a prominent loss function for multimodal feature matching. Its primary aim is to increase the dissimilarity between a sample and its associated negative samples while simultaneously reducing the distance between the sample and its positive counterpart.

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In this article, we utilize a hinge-based triplet ranking loss [29], [30], [31] to train the entire audio-visual RS retrieval network. The loss formulation is denoted as

$$\mathcal{L}(\mathcal{I},\mathcal{S}) = \sum_{\hat{\mathcal{S}}} \langle \delta - s(\mathcal{I},\mathcal{S}) + s\left(\mathcal{I},\hat{\mathcal{S}}\right)\rangle_* \oplus \sum_{\hat{\mathcal{I}}} \langle \delta - s(\mathcal{I},\mathcal{S}) + s\left(\hat{\mathcal{I}},\mathcal{S}\right)\rangle_* \quad (29)$$

where $\langle x \rangle_*$ denotes the max(x, 0) function and s denotes the cosine similarity function. Following prior practice [29], [30], [31], we set the margin parameter δ as 0.2, which defines a threshold to determine which samples are considered "sufficiently distant" negative samples. During training, this ensures that the model does not simply push all negative samples far away to minimize loss values. Instead, it focuses on modeling the semantic differences between positive and negative samples.

The hinge loss function is composed of two symmetric components. The initial summation is performed across all negative audios $\hat{\mathcal{S}}$, conditioned on the query $\mathcal{I}$. Conversely, the second component encompasses all negative images $\hat{\mathcal{I}}$, conditioned on the audio $\mathcal{S}$. To enhance computational efficiency, it is typical to sum solely over the negatives within a mini-batch during stochastic gradient descent training, instead of considering all negatives in the training set.

F. Training and Inference

The primary objective of model training is to optimize the loss function $\mathcal{L}(\mathcal{I},\mathcal{S})$ to increase the similarity between positive sample pairs while decreasing the similarity between negative sample pairs. The comprehensive network architecture, depicted in Fig. 2, illustrates the operational flow of all modules. During a single training epoch, a batch of training samples is first fed into the model, where it undergoes forward propagation and similarity computation, as described in (1) and (2), respectively. Subsequently, the loss is computed using (29). Finally, the model parameters θ_ϕ are updated based on this loss value to complete one training epoch.

During inference, unlike training, model parameters do not require updating. Therefore, it suffices to calculate the similarity of the inference batch samples based on (2). Following this, the precision value $R@K$ will be computed by sorting based on the similarity between samples.

IV. Experiments

In this section, we initially present the information of datasets and provide details on the experiments conducted. Subsequently, we compare the proposed model with existing approaches across three datasets to demonstrate the effectiveness of our approach. Next, we perform significant ablation experiments to validate the contribution of all function modules.

A. Datasets and Evaluation Metrics

We used three commonly used public datasets to carry out our experiments, including Sydney audio-visual RS dataset [10], UCM audio-visual RS dataset [10], and RSICD audio-visual RS dataset [10]. Guo et al. [10] constructed these three cross-modal datasets by expanding the audio modality from Sydney dataset [14], [38], UCM dataset [14], [39], and RSICD dataset [13]. The Sydney audio-visual RS dataset comprises 613 RS images and 3065 associated audio samples distributed across seven categories. Each image is matched with five voice samples. The UCM audio-visual RS dataset consists of 2100 RS images and 10 500 voice messages. Each image is also associated with five voice samples. The dataset is categorized into 21 classes, with every category containing 100 images and 500 voices. The RSICD audio-visual RS dataset involves 10 921 RS images and 54 605 associated voices distributed across 30 categories. Each image is linked to five voice samples. Following previous work [11], we partitioned the three datasets into training sets (60%), validation sets (20%), and test sets (20%), respectively.

In this article, we implement two tasks in RS audio-visual retrieval: 1) audio retrieval involves retrieving actual audio corresponding to the query image and 2) image retrieval entails retrieving real images that match the query audio. Specifically, only the positive pairs $(\mathcal{I}n,\mathcal{S}n)$ are referred to as ground truth. We employ two evaluation metrics, recall ($R@K$, where $K = 1$, 5, and 10), and mean recall (mR), to assess the performance of our model. $R@K$ represents the proportion of ground truth appearing in top K results [40]. Additionally, we adopt the average of all six recall values ($R@K$) proposed by Huang et al. [41] to calculate mR, providing a more comprehensive assessment of the overall model performance. If the values of $R@K$ and mR are higher, the effect of the task is better.

B. Implementation Details

Our approach is implemented by the Pytorch [42] framework. And all experimental work was conducted on NVIDIA RTX 2080Ti. Initially, we resize the RS images to 278 × 278 pixels, followed by random cropping and rotation to augment the training samples. After processing, the image is ultimately resized to 256 × 256 pixels. The input audio feature consists of 80-dimensional MFCCs extracted from a 25-ms window with a frame offset of 10 ms. The MFCC feature vectors undergo a secondary random crop, followed by normalization through cepstral mean subtraction, with no speech activity detection applied.

We use ResNet18 [32] as the visual feature extractor, and the visual embedding dimension size is 512. For the cross-modal retrieval methods we compared, we followed the comparison method mentioned in [11] and used ResNet18 to extract audio feature. Within the E-ECAPA-TDNN architecture, the dimension of the bottleneck in the SE-Block and attention modules is configured as 128. The scale dimension in Res2Block [43] is set to 8, and the total number of nodes in the fully connected layer is 512. The multihead cross-attention Λ in ICLM module uses 32-dim features and eight attention heads. The network is trained using the Adam optimizer with a batch size of 32. Following the previous practice [12], the learning rate is initially set to 2e-4 and decreases by 0.7 every 20 epochs. The margin value (δ) of the loss function is established as 0.2.

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TABLE I
COMPARISON RESULTS ON THE UCM RS AUDIO-VISUAL DATASET AND THE SYDNEY RS AUDIO-VISUAL DATASET

Method	UCM RS audio-visual dataset							Sydney RS audio-visual dataset						
	Audio Retrieval			Image Retrieval			mR	Audio Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
PVSE K=1	4.68	22.78	37.78	4.22	20.21	37.57	21.21	4.23	25.92	38.10	5.56	25.71	45.93	24.24
PVSE K=2	5.48	22.22	40.08	4.62	20.33	38.81	21.87	4.76	22.22	37.83	5.29	24.76	45.56	23.4
VSE++	10.63	36.59	54.13	10.65	39.63	60.37	35.33	8.73	28.84	43.37	8.94	35.98	57.41	30.55
CLIP	17.14	45.08	56.90	16.13	45.86	62.22	40.56	12.43	29.63	38.62	9.89	35.82	54.34	30.12
PCME	18.73	48.25	63.97	17.84	51.57	71.24	45.27	12.43	34.92	49.47	11.80	37.25	57.04	33.82
AMFMN-soft	16.51	49.44	64.76	16.75	52.33	72.52	45.37	11.64	31.75	44.18	9.37	39.47	57.04	32.24
AMFMN-sim	17.46	50.16	64.84	16.24	49.76	68.02	44.41	11.90	37.57	50.79	12.38	39.84	59.63	35.35
AMFMN-fusion	15.4	46.27	62.78	15.94	50.56	69.68	43.44	11.90	32.54	49.21	13.44	41.27	61.16	34.92
SSIA	18.49	50.79	67.14	18.68	53.82	71.34	46.71	15.07	36.77	49.47	13.59	43.01	63.33	36.87
SARCI (ours)	**22.86**	**59.29**	**72.14**	**22.38**	**57.90**	**74.00**	**51.43**	**22.22**	**48.41**	**61.90**	**16.98**	**50.63**	**69.21**	**44.89**

C. Comparison Experiments and Results

The following cross-modal retrieval methods are compared with our SARCI model.

1) PVSE [40]: PVSE combines global context with local guidance features through multihead self-attention and residual learning, calculates multiple different representations of instances, and obtains the final K embedding vector of many-to-many effects. As a control group, we employ both the $K = 2$ and $K = 1$ methods in PVSE.

2) VSE++ [31]: VSE++ explicitly introduces hard negatives in the loss of multimodal embedding without any additional mining cost.

3) CLIP [44]: CLIP is a well-known pre-trained model derived from a dataset comprising 400 million image-text pairs, which performs well on many cross-modal downstream tasks. We fine-tune it for cross-modal visual audio retrieval tasks.

4) PCME [45]: PCME utilizes probabilistic cross-modal embeddings to model extensive one-to-many correlations in image caption pairs, providing uncertainty estimates and making embeddings more interpretable.

5) AMFMN [12]: AMFMN employs MVSA modules for extracting salient features from RS images. It utilizes visual features to guide text representation, thereby enhancing cross-modal associations. Following [11], we conducted comparisons with the three methods (soft, similarity, fusion) implemented in AMFMN.

6) SSIA [11]: SSIA introduces to incorporate multiscale feature extraction and audio-driven reduction of redundant information in images.

The results of the above comparison methods and our proposed method on three public RS audio-visual cross-modal datasets are presented in Tables I and II. The results of the comparison methods are cited from [11].

1) Results on the UCM RS Audio-Visual Dataset: The results of various retrieval models on the UCM audio-visual cross-modal dataset are presented in Table I. It is evident from the results that the proposed model SARCI achieves state-of-the-art performance across all retrieval metrics. In particular, our mR index is 4.72 points higher than the previous best retrieval method SSIA. In terms of the $R@1$ score, our SARCI method demonstrates improvements of 4.37 and 3.7 points in audio retrieval and image retrieval, respectively. Noteworthy is the fact that the $R@5$ metric in audio retrieval has surged by 8.5 points, rising from 50.79 to 59.29. The great improvement of these indicators on UCM dataset strongly illustrates the excellent performance of our proposed cross-modal retrieval method SARCI. It demonstrates the overall effectiveness of the SARCI model.

TABLE II
COMPARISON RESULTS ON THE RSICD RS AUDIO-VISUAL DATASET

Method	Audio Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
PVSE K=1	0.82	1.83	2.83	0.84	3.80	6.84	2.83
PVSE K=2	0.94	2.50	3.64	0.73	3.66	7.02	3.08
VSE++	1.42	3.57	6.25	1.57	7.14	12.36	5.38
CLIP	3.44	7.38	11.25	3.31	12.19	20.06	9.61
PCME	3.41	8.45	13.09	3.18	11.70	19.45	9.9
AMFMN-soft	2.94	7.13	10.91	2.63	11.52	18.76	8.98
AMFMN-sim	3.51	7.54	11.80	3.13	11.80	19.01	9.46
AMFMN-fusion	3.69	7.73	12.13	2.66	10.46	17.39	9.01
SSIA	4.16	8.23	12.94	3.83	14.55	24.03	11.29
SARCI (ours)	**5.76**	**11.34**	**16.19**	**4.77**	**17.22**	**28.77**	**14.01**

2) Results on the Sydney RS Audio-Visual Dataset: Comparison results of different retrieval models on the Sydney audio-visual cross-modal dataset are presented in Table I. The SARCI model still greatly exceeds the previous methods in all retrieval indicators. It is worth noting that the mR index achieved by SARCI on the Sydney dataset is 8.02 points higher than the previous best retrieval method SSIA. When it comes to the $R@1$ score, our method SARCI achieves improvements of 7.15 and 3.39 points in audio retrieval and image retrieval respectively. In addition, SARCI's $R@5$ and $R@10$ retrieval indicators in audio retrieval have significantly improved by 11.64 and 12.43 points, respectively. Despite the Sydney dataset imposing challenges on the model's robustness given its relatively small scale, our proposed SARCI method

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TABLE III
ABLATION RESULTS ON THE UCM RS AUDIO-VISUAL DATASET

Ablation Model	ICLM	QDMVR	Audio Retrieval			Image Retrieval			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
a1	×	×	24.05	54.29	70.71	22.67	56.33	74.33	50.40
a2	✓	×	22.62	58.57	71.19	20.19	57.43	75.19	50.87
a3	×	✓	27.14	56.67	69.29	22.19	57.10	74.48	51.14
a4	✓	✓	22.86	59.29	72.14	22.38	57.90	74.00	**51.43**

TABLE IV
ABLATION RESULTS WITHIN QDMVR MODULE ON THE UCM RS AUDIO-VISUAL DATASET

Configuration	Audio Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
a1	24.05	54.29	70.71	22.67	56.33	74.33	50.40
a1 + MVSA [12]	24.76	59.52	68.33	21.90	55.90	72.38	50.47
a1 + QDMVR (CBAM [35])	25.95	56.67	69.76	22.00	57.38	73.33	50.85
a1 + QDMVR (high)	24.29	55.48	71.43	20.62	57.52	74.86	50.70
a1 + QDMVR (SQA)	27.14	56.67	69.29	22.19	57.10	74.48	**51.14**

significantly enhances retrieval performance. It shows the overall effectiveness of the SARCI method in dealing with audio-visual cross-modal RS retrieval task.

3) Results on the RSICD RS Audio-Visual Dataset: Results from Table II provide a comparative overview of different retrieval models on the RSICD audio-visual cross-modal dataset. Despite its large scale, the RSICD dataset exhibits limitations in fine-grained details for the RS cross-modal retrieval tasks [12]. A drawback of the RSICD dataset is the presence of annotated repetitions, leading to numerous audio descriptions for similar scenes being uniform rather than providing more nuanced details [12]. The limitation of low granularity has led to comparatively inferior performance in cross-modal retrieval tasks on the RSICD dataset compared to other datasets. Nevertheless, our proposed SARCI model has enhanced the retrieval performance on the RSICD dataset.

For the RSICD dataset, SARCI consistently demonstrates leading performance across all evaluation metrics. Especially, our mR index is 2.72 points higher than the previous best retrieval method SSIA. When it comes to the $R@1$ score, our method SARCI achieves improvements of 1.6 and 0.94 points in audio retrieval and image retrieval respectively. Moreover, the $R@5$ metrics in audio retrieval and image retrieval have increased 3.11 and 2.67 points, respectively, for RSICD dataset. And the $R@10$ metrics in two retrieval types have correspondingly improved by 3.25 and 4.74 points. SARCI's outstanding performance on the challenging large-scale RSICD dataset demonstrates its versatility in handling significant issues.

D. Ablation Study

As shown in Table III, we conduct five corresponding localization experiments to investigate the efficacy of each module in our approach.

1) a1: We combine the visual feature extractor ResNet18, audio feature extractor E-ECAPA-TDNN, and the hinge-based triplet ranking loss as our baseline model. The cosine similarity is measured by encoded features of the visual backbone ResNet18 and audio backbone E-ECAPA-TDNN.

2) a2: Sequentially, a2 is to demonstrate the efficient interaction of the ICLM module for cross-modal features.

3) a3: Model a3 is used to explore the QDMVR module's processing capabilities for tackling issues of multiscale and redundant information.

4) a4 (SARCI): All function modules proposed are included in a4, which serves as an important ablation contrast model.

Table III represents all the ablation retrieval results on the UCM audio-visual dataset.

1) Comparing model a1 with a2, the ICLM module has improved the mR metric from 50.40 to 50.87. In particular, the $R@5$ indicator in audio retrieval has increased by 4.28 points. The improvement in retrieval performance strongly indicates that the ICLM module has effectively engaged in meaningful cross-modal interactions.

2) When it comes to model a3 in comparison with a1, the retrieval capability has been greatly enhanced by the QDMVR module. QDMVR has promoted the increase of the mR indicator from 50.40 to 51.14. It is worth mentioning that the $R@1$ metric of audio retrieval has been improved by 3.09 points. It illustrates that QDMVR effectively optimizes visual features by coping with the multiscale and target redundant characteristics of RS images.

3) Seen from model a2 to a4 or a3 to a4, the collaborative effect of QDMVR and ICLM in audio-visual cross-modal RS retrieval task has been demonstrated. In general, SARCI is 1.03 points higher in the mR metric than the baseline model.

To explore the relative effectiveness of the SQA mechanism, we conduct comparison experiments with CBAM attention, shown in Table IV. Within Table IV, a1 + QDMVR (SQA)

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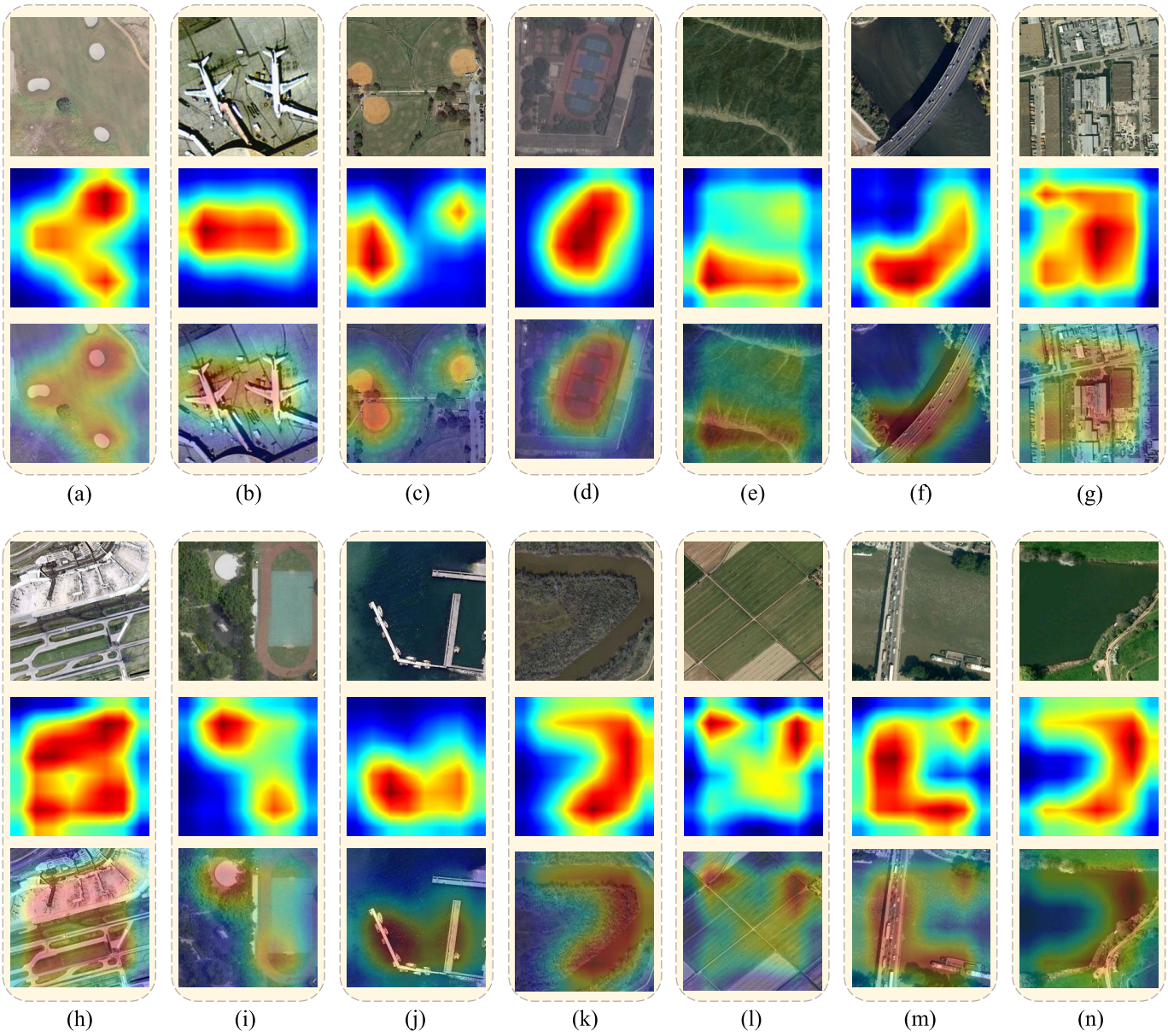


Fig. 4. Semantic visualization of learned visual feature output by QDMVR module for RS images. We present feature semantic visualization results of image samples from various RS scenes, totaling 14 [from (a) to (n)]. (Top) Input RS image. (Center) Attention map of visual feature. (Bottom) Overlaid image.

refers to model a1 plus QDMVR with our proposed SQA mechanism. Whereas, a1 + QDMVR (CBAM [35]) refers to replace the SQA with CBAM for comparison. Apparently, our SQA mechanism, which works in QDMVR, exceeds the CBAM quantitatively. It indicates that our SQA mechanism has superiorly optimized visual feature. The symmetric quaternary structure and the fusion mode of spatial and channel attention both matter for the SQA mechanism.

And a1 + MVSA [12] refers to replace the QDMVR with MVSA [12] for comparison. We observe that our proposed QDMVR module achieves superior performance compared to the MVSA module. This improvement is attributed to the adaptive learning of redundant information across various visual scales, focusing on more meaningful semantic information.

To explore the effectiveness of our multilevel feature fusion, a1 + QDMVR (high) in Table IV represents a comparison model to use the optimized high-level visual feature F_h' as the visual representation instead of visual feature representation V_g in our designed. It can be observed that the performance of this model is 0.44 lower than that of model a1 + QDMVR (SQA) in terms of the mR metric. Overall, our QDMVR model demonstrates better retrieval performance, affirming the effectiveness of incorporating adaptive learning of visual features containing small scales.

E. Semantic Visualization of Visual Feature for RS Images

In this section, we analyze the attention map output by the QDMVR module to explore its understanding of RS images. The semantic visualization results are shown in Fig. 4.

In numerous RS scenarios, it is evident that the model effectively captures meaningful and salient semantic information. For instance, the examples (a)–(d), (f), (i), and (m) demonstrate

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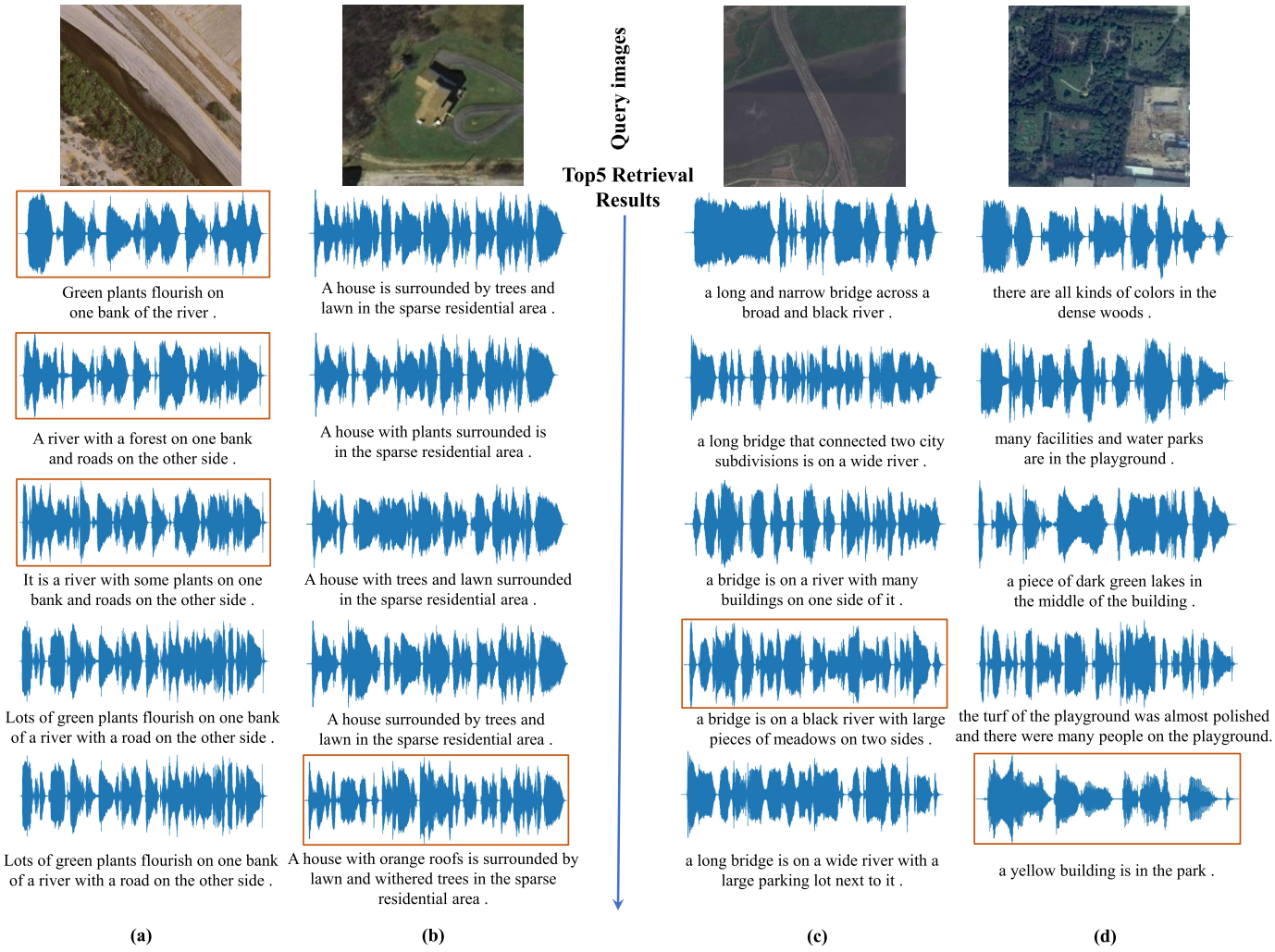


Fig. 5. Examples of the retrieval results in audio retrieval. (Top) Query RS images. (Bottom) Corresponding retrieval results for each RS image. The red box represents the correct results returned. Example (a) and (b) are from UCM dataset. While (c) and (d) show the results in RSICD dataset.

that the model can significantly focus on target objects, such as airplanes, baseball fields, playgrounds, bridges, and rivers. Regarding example (h), due to phrases like "many runways sit beside the terminal building and a plane is running on one of it" in the corresponding audio, the cross-modal interaction within the ICLM module can effectively guide visual features. Notably, salient semantic entities such as "runways," "terminal," and "plane" contribute significantly to this process. The attention maps distinctly reveal the model's focus on airport runways and adjacent terminal structures. Through examples (f) and (m), we can find that the model not only attends to bridges but also captures semantic features on both riverbanks, such as vegetation and boats. Texture information of the mountains is also captured in example (e). These examples illustrate that our QDMVR module effectively focuses on the key semantics within visual information.

F. Visualization of Retrieval Results

In Figs. 5 and 6, we present some retrieval results of the SARCI model in audio retrieval and image retrieval, respectively.

In Fig. 5, the results displayed showcase the remarkable performance of the SARCI model in audio retrieval. We find that the correct results are ranked higher in the retrieval results. Moreover, the model correctly captures some RS semantic objects such as "plants," "roads," "houses," and "rivers". Meanwhile, in the RSICD dataset, the top retrieval results still include relevant information from the visual input. Despite the issue of less fine-grained annotations in the RSICD dataset, our model is still able to retrieve content relevant to the input, with many returns belonging to the relative category.

As shown in Fig. 6, the correct image retrieval results for the samples are ranked in the top 2 in the UCM RS audio-visual dataset. We observe that the results from other retrievals fall into the same category as the result images, indicating that the model has learned associations between paired modalities to some extent. The sample results in the RSICD dataset exhibit a similar situation, with the top five returned results being even more similar. From the retrieval results, we find that the model has learned useful semantics from audio modality. For instance, the descriptive object, such as "curved river," has been understood by the model, as shown

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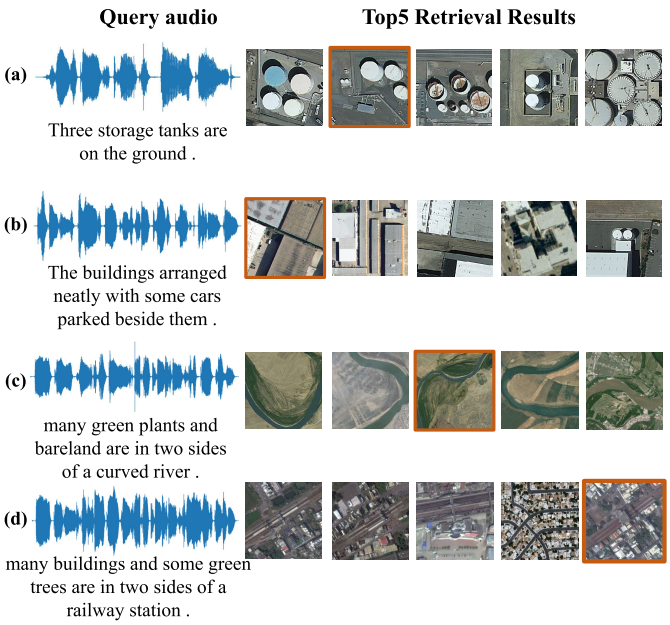


Fig. 6. Examples of the retrieval results in image retrieval. (Left) Query audio. (Right) Corresponding retrieval results for each piece of audio. The red box represents the correct result returned. Example (a) and (b) are from UCM dataset. While (c) and (d) show the results in RSICD dataset.

in Example (c) of Fig. 6. However, some detailed cross-modal semantic relationships are not well recognized from the observation of Example (a) in Fig. 6. The top images returned in Example (a) show that the model did not fully grasp the precise semantics of "three storage tanks," learning the semantics of target entity but ignoring the numerical semantics.

V. Conclusion

In this article, we introduce a novel SARCI network designed for RS audio-visual cross-modal retrieval. To better handle the multiscale found in RS images and alleviate redundant information, we propose the QDMVR module within the visual branch to explore salient RS vision information across different scales. To facilitate more meaningful interaction among information from diverse modalities, we introduce the ICLM for cross-modal feature interaction. The ICLM effectively addresses expression variations between cross-modal features, thereby improving the performance of the retrieval model. Our model exhibits superior performance across three publicly available RS cross-modal datasets, including the Sydney, UCM, and RSICD audio-visual datasets. Indeed, there is a potential for improving cross-modal retrieval performance. Future work may investigate how to optimize the objective function of models to learn more comprehensive cross-modal correspondence. The RS audio-visual cross-modal retrieval method may enhance the efficiency of recognizing and analyzing surface features, environmental changes, animal behaviors, and other information by learning the correlation between image and audio data. Subsequent work will expand the application of this technology in practice.

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